The Cat Righting Reflex: Full-Cylinder and Half-Cylinder Models with No-Wringing Constraint

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Abstract

We analyze the cat righting reflex using two models: (I) two identical full cylinders joined at a fixed bend angle θ, and (II) two identical half-cylinders (the belly halves) with a center-of-mass offset from the geometric axis. Working entirely in Cartesian coordinates with the origin at the system center of mass, we compute the total angular momentum by explicit integration over mass elements. The no-wringing constraint—equal and opposite spin rates about outward-pointing axes—determines a unique whole-body counter-rotation (the righting flip) that maintains zero total angular momentum. We derive the gearing ratio for each model, specialize to $\theta = 90°$, and show that the half-cylinder gearing is suppressed by the factor $1-32/(9\pi^2) \approx 0.640$, equal to the ratio of axial moments of inertia. We then derive the internal torques at the joint in both cases, decomposing each into spin, orbital, and whole-body-rotation contributions. Every formula is verified by independent SymPy symbolic integration. Detailed calculations are provided in the appendices so that the reader may check every step.

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1 Introduction

A cat released upside-down in free fall can right itself and land on its feet, even with zero initial angular momentum. The simplest mechanical model replaces the cat with two rigid cylinders joined at a fixed bend angle, each spinning about its own axis in opposite senses (the "no-wringing" constraint). Because the two spin axes are non-parallel, the spin angular momenta do not cancel vectorially, and a compensating whole-body rotation about the arch axis is required to maintain $\boldsymbol{L} = \mathbf{0}$. This compensating rotation is the righting flip.

We treat two variants of this model: full cylinders (Section 3) and half-cylinders (Section 4). The half-cylinder model represents a cat whose cross-section is semicircular (the belly side only), introducing a center-of-mass offset $\delta = 4R/(3\pi)$ from the geometric axis. We compare the gearing ratios (Section 5), derive the internal torques (Section 6), and present complete step-by-step calculations in the appendices.

2 Common Setup

2.1 Coordinate system

The origin is at the system center of mass. The y-axis points along the symmetry axis of the arch (upward, from the system center of mass toward the joint), x is horizontal (the flip axis), and z completes the right-handed system. The half-angle is $\alpha \equiv \theta/2$. The outward-pointing axis unit vectors are

$$\hat{\boldsymbol{n}}_f = (\sin\alpha,\ \cos\alpha,\ 0), \qquad \hat{\boldsymbol{n}}_r = (-\sin\alpha,\ \cos\alpha,\ 0). \tag{1}$$

2.2 The no-wringing constraint

The front cylinder spins at $+\omega$ about $\hat{\boldsymbol{n}}_f$ and the rear at $-\omega$ about $\hat{\boldsymbol{n}}_r$. This ensures material continuity at the joint—no relative twist.

2.3 Perpendicular basis vectors

For each cylinder we need two directions perpendicular to the axis:

$$\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{z}} = (0, 0, 1), \tag{2}$$

$$\hat{\boldsymbol{e}}_{2f} = \hat{\boldsymbol{n}}_f \times \hat{\boldsymbol{z}} = (\cos\alpha, -\sin\alpha, 0), \tag{3}$$

$$\hat{\boldsymbol{e}}_{2r} = \hat{\boldsymbol{n}}_r \times \hat{\boldsymbol{z}} = (\cos\alpha, \sin\alpha, 0). \tag{4}$$

The cross-product relations are:

$$\hat{\boldsymbol{n}}_f \times \hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}_{2f}, \qquad \hat{\boldsymbol{n}}_f \times \hat{\boldsymbol{e}}_{2f} = -\hat{\boldsymbol{e}}_1. \tag{5}$$

3 Model I: Full Cylinders

3.1 Geometry

Two identical uniform cylinders of mass m, length $\ell = s_2 - s_1$, and radius R are joined at a bend with fixed opening angle θ. Each cylinder is parameterized by arc-length s from s_1 to s_2 measured from the joint; the gap $s_1 = R/\tan\alpha$ prevents overlap. (See Figure 1.)

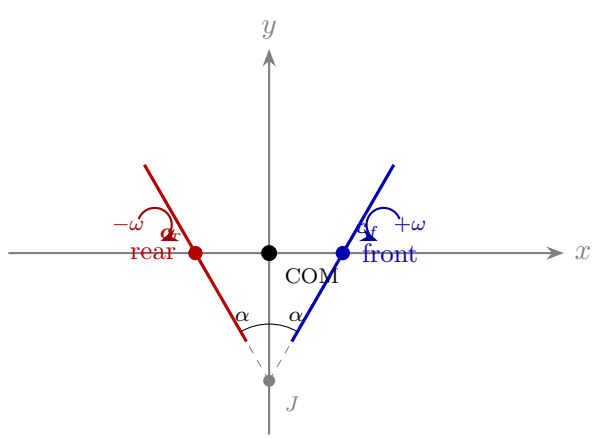

Figure 1: Full-cylinder model. Both cylinder centers of mass lie on the x-axis.

The center of mass of each cylinder is at distance $\bar{s} = (s_1 + s_2)/2$ along its axis from the joint. Setting the origin at the system center of mass:

$$J = (0, -\bar{s}\cos\alpha, 0), \quad \boldsymbol{c}_f = (+\bar{s}\sin\alpha, 0, 0), \quad \boldsymbol{c}_r = (-\bar{s}\sin\alpha, 0, 0). \tag{6}$$

Both cylinder centers lie on the x-axis.

3.2 Angular momentum

A mass element at position $\boldsymbol{p} = J + s\,\hat{\boldsymbol{n}}_f + \rho\,\hat{\boldsymbol{e}}_\perp(\phi)$ in the front cylinder (with $\hat{\boldsymbol{e}}_\perp = \cos\phi\,\hat{\boldsymbol{e}}_{2f} + \sin\phi\,\hat{\boldsymbol{e}}_1$, $\phi \in [0, 2\pi)$) has velocity $\boldsymbol{v} = \boldsymbol{v}_{\text{spin}} + \boldsymbol{\Omega} \times \boldsymbol{p}$. The spin velocity is $\boldsymbol{v}_{\text{spin}} = \omega\rho(\sin\phi\,\hat{\boldsymbol{e}}_{2f} - \cos\phi\,\hat{\boldsymbol{e}}_1)$.

The complete integration (Appendix A) yields, with the no-wringing constraint:

$$L_x = \frac{m}{6}\left[\Omega_x\,\mathcal{I}_{xx}^{(\text{full})} + 6R^2\omega\sin\alpha\right], \tag{7}$$

$$L_y = \frac{m}{6}\,\Omega_y\,\mathcal{I}_{yy}^{(\text{full})}, \tag{8}$$

$$L_z = \frac{m}{6}\,\Omega_z\,\mathcal{I}_{zz}^{(\text{full})}, \tag{9}$$

where

$$\mathcal{I}_{xx}^{(\text{full})} = (\ell^2 - 3R^2)\cos^2\alpha + 6R^2, \tag{10}$$

$$\mathcal{I}_{yy}^{(\text{full})} = (\ell^2 + 12\bar{s}^2 - 3R^2)\sin^2\alpha + 6R^2, \tag{11}$$

$$\mathcal{I}_{zz}^{(\text{full})} = \ell^2 + 12\bar{s}^2\sin^2\alpha + 3R^2. \tag{12}$$

The spin term appears *only* in L_x, and $\mathcal{I}_{xx}^{(\text{full})}$ has no $\bar{s}$-dependence (the cylinder centers lie on the x-axis).

3.3 Gearing ratio

Setting $\boldsymbol{L} = \boldsymbol{0}$: $\Omega_y = \Omega_z = 0$ and

$$\boxed{\left.\frac{\Omega_x}{\omega}\right|_{\text{full}} = -\frac{6R^2\sin\alpha}{(\ell^2 - 3R^2)\cos^2\alpha + 6R^2}\,.} \tag{13}$$

4 Model II: Half-Cylinders

4.1 Geometry

Each half-cylinder is the top half ($\phi \in [0, \pi]$) of radius R, length L, mass m. Each extends outward from the joint ($s \in [0, L]$). In addition to the perpendicular basis $(\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_{2f})$, we define the "belly" direction (from the geometric axis toward the half-disk center of mass):

$$\hat{\boldsymbol{b}}_f = (\cos\alpha, -\sin\alpha, 0), \qquad \hat{\boldsymbol{b}}_r = (-\cos\alpha, -\sin\alpha, 0). \tag{14}$$

Note $\hat{\boldsymbol{b}}_f = \hat{\boldsymbol{e}}_{2f}$ and the out-of-plane direction $\hat{\boldsymbol{e}}_f = \hat{\boldsymbol{b}}_f \times \hat{\boldsymbol{n}}_f = \hat{\boldsymbol{z}}$.

4.2 Center-of-mass offset

The half-disk center of mass is offset from the geometric center by

$$\delta = \frac{4R}{3\pi}\,. \tag{15}$$

(Derivation: $\delta = \frac{1}{\pi R^2/2}\int_0^\pi\int_0^R \rho\sin\phi\;\rho\,d\rho\,d\phi = \frac{4R}{3\pi}$.)

The half-cylinder center of mass from the joint is $\boldsymbol{r}_f^{\rm com} = (L/2)\hat{\boldsymbol{n}}_f + \delta\hat{\boldsymbol{b}}_f$. The system center of mass is at $y_{\rm com} = (L/2)\cos\alpha - \delta\sin\alpha$ above the joint, and both half-cylinder centers lie on the x-axis: $\boldsymbol{c}_f = (+d_x, 0, 0)$, $\boldsymbol{c}_r = (-d_x, 0, 0)$ with $d_x = (L/2)\sin\alpha + \delta\cos\alpha$.

4.3 Inertia tensor

The body-frame inertia tensor about the center of mass is diagonal with principal moments (Appendix B):

$$I_{ee} = m\left(\frac{L^2}{12} + \frac{(9\pi^2-64)R^2}{36\pi^2}\right), \tag{16}$$

$$I_{bb} = \frac{m(L^2+3R^2)}{12}\,, \tag{17}$$

$$I_{nn} = \frac{(9\pi^2-32)mR^2}{18\pi^2}\,. \tag{18}$$

The ratio $I_{nn}/I_{\rm ax}^{\rm (full)} = 1 - 32/(9\pi^2) \approx 0.640$.

4.4 Angular momentum

The integration (Appendix C) with $s \in [0, L]$, $\phi \in [0, \pi]$, and the no-wringing constraint gives $L_y = L_z = 0$ and

$$L_x = A\,\omega + B\,\Omega_x\,, \tag{19}$$

where

$$A = \frac{(9\pi^2-32)mR^2\sin\alpha}{9\pi^2} = 2I_{nn}\sin\alpha, \tag{20}$$

$$B = \frac{m}{18\pi^2}\big[3\pi^2L^2\cos^2\alpha + R^2(9\pi^2 + (9\pi^2-64)\sin^2\alpha)\big]. \tag{21}$$

4.5 Gearing ratio

$$\boxed{\left.\frac{\Omega_x}{\omega}\right|_{\rm half} = \frac{-2(9\pi^2-32)R^2\sin\alpha}{3\pi^2L^2\cos^2\alpha + R^2(9\pi^2+(9\pi^2-64)\sin^2\alpha)}\,.} \tag{22}$$

5 Comparison of the Two Models

5.1 The $\theta = 90°$ case

Full cylinder: $\Omega_x/\omega = -6\sqrt{2}\,R^2/(\ell^2+9R^2)$.
Half-cylinder: $\Omega_x/\omega = -\sqrt{2}(9\pi^2-32)R^2/(\frac{3}{2}\pi^2L^2 + \frac{1}{2}R^2(27\pi^2-64))$.

At $L/R = 4$: full gives -0.339, half gives -0.238.

5.2 Long-cylinder limit

Both scale as $(R/L)^2$. The ratio of leading coefficients is

$$\frac{(\Omega/\omega)_{\text{half}}}{(\Omega/\omega)_{\text{full}}} \xrightarrow{L\gg R} 1 - \frac{32}{9\pi^2} \approx 0.640 = \frac{I_{nn}}{I_{\text{ax}}^{(\text{full})}} \,. \tag{23}$$

5.3 Comparison table

Table 1: Gearing ratio Ω/ω at $L/R = 4$. The last column gives the spin angle for a 180° flip (half/full, in degrees).

θ	Half-cyl	Full-cyl	Ratio	ϕ_{180} (half/full)
30	−0.055	−0.086	0.64	3260/2101
45	−0.088	−0.134	0.65	2057/1340
60	−0.126	−0.191	0.66	1427/945
90	−0.238	−0.339	0.70	757/530
120	−0.436	−0.562	0.78	413/320
150	−0.764	−0.844	0.91	236/213

6 Internal Torques

The total angular momentum is zero, but each *half* of the system carries nonzero (and time-varying) angular momentum. The rate of change $d\boldsymbol{L}_f/dt$ equals the net torque on the front half, which in free fall is entirely the internal joint torque:

$$\boldsymbol{\tau}_{\text{front}} = \frac{d\boldsymbol{L}_f}{dt} \,, \qquad \boldsymbol{\tau}_{\text{rear}} = -\boldsymbol{\tau}_{\text{front}} \,. \tag{24}$$

6.1 Decomposition

The angular momentum of each half has three contributions:

1. **Spin:** angular momentum from rotation of the half-cylinder about its geometric axis, $\boldsymbol{L}_{\text{spin}} = I_{\text{ax}}\,\omega\,\hat{\boldsymbol{n}}_f$ (where I_{ax} is $mR^2/2$ for the full cylinder or I_{nn} for the half-cylinder).

2. **Orbital:** angular momentum from the half-cylinder's center of mass moving about the system center of mass. For the full cylinder ($\delta = 0$), the center of mass lies on the geometric axis and the spin does not move it, so this term is zero. For the half-cylinder, the offset center of mass orbits the geometric axis at rate ω, giving $\boldsymbol{L}_{\text{orb}} = m\,\boldsymbol{c}_f \times \boldsymbol{v}_{\text{com}}^{\text{spin}}$.

3. **Whole-body rotation:** angular momentum from the entire system rotating at Ω_x about $\hat{\boldsymbol{x}}$. This is $\boldsymbol{L}_{\text{rot}} = \mathsf{I}_f^{(\text{lab})} \cdot \Omega_x\,\hat{\boldsymbol{x}}$, where $\mathsf{I}_f^{(\text{lab})}$ is the lab-frame inertia tensor of the front half about the system center of mass.

Since $\Omega_x = G\,\omega$ with G the gearing ratio, all three contributions are proportional to ω, and their time derivatives are proportional to $\dot{\omega}$.

6.2 Full-cylinder torques

For the full cylinder, $\delta = 0$ so the orbital term vanishes. The angular momentum of the front half is (Appendix D):

$$\boldsymbol{L}_f^{(\text{full})} = \big(0,\ C^{(\text{full})}\omega + D^{(\text{full})}\Omega_x,\ 0\big), \tag{25}$$

where (from the integration in Appendix A):

$$C^{(\text{full})} = \tfrac{1}{2} m R^2 \cos\alpha, \tag{26}$$

$$D^{(\text{full})} = \frac{m}{24}\sin(2\alpha)(3R^2 - \ell^2). \tag{27}$$

The x-component of $\boldsymbol{L}_f$ is identically zero (verified in Appendix D), reflecting the fact that each half independently has $L_x = 0$. The torque on the front half is

$$\boldsymbol{\tau}_f^{(\text{full})} = \big(C^{(\text{full})} + D^{(\text{full})}G^{(\text{full})}\big)\dot{\omega}\ \hat{\boldsymbol{y}}\,. \tag{28}$$

This decomposes into:

- $C^{(\text{full})}\dot{\omega}\,\hat{\boldsymbol{y}}$: the y-component of $d(I_{\text{ax}}\omega\hat{\boldsymbol{n}}_f)/dt = I_{\text{ax}}\dot{\omega}\hat{\boldsymbol{n}}_f$, which contributes $I_{\text{ax}}\cos\alpha\,\dot{\omega}$ in y and $I_{\text{ax}}\sin\alpha\,\dot{\omega}$ in x.
- $D^{(\text{full})}G^{(\text{full})}\dot{\omega}\,\hat{\boldsymbol{y}}$: the y-component of $d(\mathsf{I}_f\cdot\Omega_x\hat{\boldsymbol{x}})/dt$, i.e., $I_{xy}^{(\text{lab})}\,G\,\dot{\omega}$. Its x-component, $I_{xx}^{(\text{lab})}\,G\,\dot{\omega}$, exactly cancels the x-component of the spin torque, confirming $L_x^{(f)} = 0$.

6.3 Half-cylinder torques

For the half-cylinder, all three terms contribute (Appendix E):

$$\boldsymbol{\tau}_f^{(\text{half})} = \big(C^{(\text{half})} + D^{(\text{half})}G^{(\text{half})}\big)\dot{\omega}\ \hat{\boldsymbol{y}}\,, \tag{29}$$

where

$$C^{(\text{half})} = \frac{Rm(4L\sin\alpha + 3\pi R\cos\alpha)}{6\pi}\,, \tag{30}$$

$$D^{(\text{half})} = \frac{m(-64R^2 + 3\pi^2(-L^2 + 3R^2))\sin\theta}{72\pi^2}\,. \tag{31}$$

The coefficient $C^{(\text{half})}$ contains two pieces:

1. $I_{nn}\cos\alpha\,\dot{\omega}$: the y-component of the spin torque (analogous to the full-cylinder case but with I_{nn} instead of I_{ax}).
2. $m\,d_x\,\delta\,\dot{\omega}$: the orbital torque from the accelerating center-of-mass wobble. **This term is absent in the full-cylinder case** ($\delta = 0$).

6.4 Numerical comparison

At $R = 1$, $L = 4$, $\theta = 90°$, $m = 1$:

Table 2: Torque contributions (per unit $\dot{\omega}$) for each model.

Contribution	Full cylinder	Half-cylinder
Spin (y-comp)	0.354	0.226
Orbital	0	0.728
Whole-body rotation (y-comp)	0.154	0.150
Total τ_y	0.508	1.104

The half-cylinder joint torque is more than twice the full-cylinder value because of the large orbital contribution. This orbital term arises entirely from the center-of-mass offset δ.

6.5 Newton's third law and energy

The torque on the rear half is $-\boldsymbol{\tau}_f$, satisfying Newton's third law. Both torques are along $\hat{\boldsymbol{y}}$: there is no out-of-plane (z) component and no bending component (which would be along $\hat{\boldsymbol{z}}$ about the joint).

The cat's muscles do work at the instantaneous rate $P = \tau_y \cos\alpha \cdot \omega$ (the component of the torque along the geometric axis times the spin rate). For a time-symmetric spin profile $\omega(t) = -\omega(T-t)$, the net work over a complete cycle is zero: the cat borrows kinetic energy from its muscles and returns it, ending at rest but flipped.

7 Where Intuition Goes Wrong

"You can't add angular momenta about different axes." Angular momentum is a vector. The sum of $I_{\text{ax}}\omega\hat{\boldsymbol{n}}_f$ and $-I_{\text{ax}}\omega\hat{\boldsymbol{n}}_r$ is a perfectly well-defined vector along $+\hat{\boldsymbol{x}}$.

"The bend angle must change." $\boldsymbol{L} = \boldsymbol{0}$ is satisfied at every instant with fixed θ and constant spin and flip rates.

"Geometric phase is the explanation." The Berry phase picture is valid but unnecessary. The mechanism is a direct angular momentum balance: non-parallel axes produce a net L_x that must be canceled by whole-body rotation.

8 Conclusion

The two-cylinder cat righting model admits a clean closed-form solution for both full and half-cylinder cross-sections. The only difference between the two models is the axial moment of inertia: the half-cylinder has $I_{nn} = (1 - 32/(9\pi^2))\, mR^2/2 \approx 0.640\, I_{\text{ax}}^{(\text{full})}$, reducing the gearing by the same factor in the long-cylinder limit. The center-of-mass offset introduces a large orbital contribution to the internal joint torque but does not affect the gearing ratio's functional form. All results are verified by symbolic integration.

A Full-Cylinder Angular Momentum Integration

We compute the angular momentum of the front cylinder in full detail. The rear follows by $\sin\alpha \to -\sin\alpha$, $\omega \to -\omega$.

A.1 Position and velocity

Position from the system center of mass:

$$p_x = s\sin\alpha + \rho\cos\phi\cos\alpha, \tag{32}$$
$$p_y = (s-\bar{s})\cos\alpha - \rho\cos\phi\sin\alpha, \tag{33}$$
$$p_z = \rho\sin\phi. \tag{34}$$

Spin velocity:

$$\boldsymbol{v}_{\text{spin}} = \omega\rho\begin{pmatrix}\sin\phi\cos\alpha\\ -\sin\phi\sin\alpha\\ -\cos\phi\end{pmatrix}. \tag{35}$$

Whole-body velocity: $\boldsymbol{v}_\Omega = \boldsymbol{\Omega}\times\boldsymbol{p}$ with general $\boldsymbol{\Omega} = (\Omega_x, \Omega_y, \Omega_z)$.

A.2 Spin–position cross product

$$(\boldsymbol{p}\times\boldsymbol{v}_{\text{spin}})_x = \omega\rho\cos\phi(\bar{s}-s)\cos\alpha + \omega\rho^2\sin\alpha, \tag{36}$$
$$(\boldsymbol{p}\times\boldsymbol{v}_{\text{spin}})_y = \omega\rho^2\cos\alpha + \omega\rho s\sin\alpha\cos\phi, \tag{37}$$
$$(\boldsymbol{p}\times\boldsymbol{v}_{\text{spin}})_z = -\omega\rho\sin\phi[s-\bar{s}\cos^2\alpha]. \tag{38}$$

Verification of (36)*:*

$$\begin{aligned} p_y v_z^{\text{spin}} - p_z v_y^{\text{spin}} &= [(s-\bar{s})\cos\alpha - \rho\cos\phi\sin\alpha](-\omega\rho\cos\phi) - \rho\sin\phi(-\omega\rho\sin\phi\sin\alpha)\\ &= -\omega\rho\cos\phi(s-\bar{s})\cos\alpha + \omega\rho^2\cos^2\phi\sin\alpha + \omega\rho^2\sin^2\phi\sin\alpha\\ &= \omega\rho\cos\phi(\bar{s}-s)\cos\alpha + \omega\rho^2\sin\alpha. \checkmark \end{aligned}$$

Verification of (37)*:*

$$\begin{aligned} p_z v_x^{\text{spin}} - p_x v_z^{\text{spin}} &= \rho\sin\phi\cdot\omega\rho\sin\phi\cos\alpha - (s\sin\alpha + \rho\cos\phi\cos\alpha)(-\omega\rho\cos\phi)\\ &= \omega\rho^2\sin^2\phi\cos\alpha + \omega\rho\cos\phi\, s\sin\alpha + \omega\rho^2\cos^2\phi\cos\alpha\\ &= \omega\rho^2\cos\alpha + \omega\rho s\sin\alpha\cos\phi. \checkmark \end{aligned}$$

Verification of (38)*:*

$$\begin{aligned} p_x v_y^{\text{spin}} - p_y v_x^{\text{spin}} &= (s\sin\alpha + \rho\cos\phi\cos\alpha)(-\omega\rho\sin\phi\sin\alpha)\\ &\quad - [(s-\bar{s})\cos\alpha - \rho\cos\phi\sin\alpha](\omega\rho\sin\phi\cos\alpha)\\ &= -\omega\rho\sin\phi[s\sin^2\alpha + \rho\cos\phi\sin\alpha\cos\alpha\\ &\qquad + (s-\bar{s})\cos^2\alpha - \rho\cos\phi\sin\alpha\cos\alpha]\\ &= -\omega\rho\sin\phi[s-\bar{s}\cos^2\alpha]. \checkmark \end{aligned}$$

A.3 Integration over ϕ

With $\int_0^{2\pi}\cos\phi\,d\phi = \int_0^{2\pi}\sin\phi\,d\phi = 0$:

From (36): the $\cos\phi$ term vanishes, leaving $\int_0^{2\pi}(\cdots)_x\,\rho\,d\phi = 2\pi\omega\rho^3\sin\alpha$.

From (37): the $\cos\phi$ term vanishes, leaving $\int_0^{2\pi}(\cdots)_y\,\rho\,d\phi = 2\pi\omega\rho^3\cos\alpha$.

From (38): entire expression $\propto \sin\phi$, so $\int_0^{2\pi}(\cdots)_z\,\rho\,d\phi = 0$.

A.4 Orbit–position cross product

Using $\boldsymbol{p} \times (\boldsymbol{\Omega} \times \boldsymbol{p}) = |\boldsymbol{p}|^2\boldsymbol{\Omega} - (\boldsymbol{p} \cdot \boldsymbol{\Omega})\boldsymbol{p}$ and integrating over ϕ (only $\cos^2\phi$ and $\sin^2\phi$ survive, each giving π), we obtain expressions involving Ω_x, Ω_y, Ω_z multiplied by combinations of s^2, $(\bar{s} - s)^2$, and ρ^2 terms. The needed s-integrals are:

$$\int_{s_1}^{s_2} (\bar{s} - s)^2 \, ds = \frac{\ell^3}{12}, \quad \int_{s_1}^{s_2} s^2 \, ds = \bar{s}^2\ell + \frac{\ell^3}{12}, \quad \int_{s_1}^{s_2} s(s - \bar{s}) \, ds = \frac{\ell^3}{12}. \tag{39}$$

After collecting all terms (the detailed algebra is in Section A.4.1 below), the front cylinder's total angular momentum is:

$$(L_f)_x = \frac{m}{24}\big[\Omega_x(2\ell^2\cos^2\alpha - 6R^2\cos^2\alpha + 12R^2) + \Omega_y \sin(2\alpha)(3R^2 - \ell^2) + 12R^2\omega\sin\alpha\big], \tag{40}$$

$$(L_f)_y = \frac{m}{24}\big[\Omega_x \sin(2\alpha)(3R^2 - \ell^2) + \Omega_y(2\ell^2\sin^2\alpha + 24\bar{s}^2\sin^2\alpha - 6R^2\sin^2\alpha + 12R^2) + 12R^2\omega\cos\alpha\big], \tag{41}$$

$$(L_f)_z = \frac{m}{12}\Omega_z(\ell^2 + 12\bar{s}^2\sin^2\alpha + 3R^2). \tag{42}$$

Adding the rear cylinder (with $\sin\alpha \to -\sin\alpha$ and $\omega \to -\omega$), the Ω_y–Ω_x cross terms cancel and the spin $\cos\alpha$ terms cancel, yielding equations (7)–(9).

A.4.1 Orbit-term detail

After ϕ-integration, the x-component of the orbit term for the front cylinder is:

$$\int_0^{2\pi} [\boldsymbol{p}\times(\boldsymbol{\Omega}\times\boldsymbol{p})]_x \, \rho \, d\phi = 2\pi\rho\big\{\Omega_x[(\bar{s}-s)^2\cos^2\alpha + \tfrac{1}{2}\rho^2\sin^2\alpha + \rho^2] - \Omega_y \sin\alpha\cos\alpha[s(s-\bar{s}) - \tfrac{1}{2}\rho^2]\big\}. \tag{43}$$

Integrating over $\rho \in [0, R]$ and $s \in [s_1, s_2]$ with $dm = \frac{m}{\pi R^2 \ell}\rho \, d\rho \, d\phi \, ds$:

$$(L_f^{\text{orb}})_x = m\big\{\Omega_x\big[\tfrac{\ell^2}{12}\cos^2\alpha + \tfrac{R^2}{4}\sin^2\alpha + \tfrac{R^2}{2}\big] - \Omega_y \sin\alpha\cos\alpha\big[\tfrac{\ell^2}{12} - \tfrac{R^2}{4}\big]\big\}. \tag{44}$$

Adding the spin contribution $(mR^2/2)\omega\sin\alpha$ gives (40) after combining $\frac{R^2}{4}\sin^2\alpha + \frac{R^2}{2} = \frac{R^2}{4}(2 + \sin^2\alpha) = \frac{R^2}{4}(3 - \cos^2\alpha)$.

B Half-Cylinder Inertia Tensor

Body-frame coordinates relative to the center of mass: $x_b = \rho\cos\phi$, $y_b = \rho\sin\phi - \delta$, $z_b = s$ with $\phi \in [0, \pi]$, $\rho \in [0, R]$, $s \in [-L/2, L/2]$, and density $\rho_0 = 2m/(\pi R^2 L)$.

B.1 $I_{ee} = \int (y_b^2 + z_b^2) \, dm$

$$I_{ee} = \rho_0 \int_{-L/2}^{L/2} \int_0^R \int_0^\pi [(\rho\sin\phi - \delta)^2 + s^2] \, \rho \, d\phi \, d\rho \, ds \tag{45}$$

$$= \rho_0 \left[\int s^2 \, ds \cdot \iint \rho \, d\phi \, d\rho + \int ds \cdot \iint (\rho\sin\phi - \delta)^2 \rho \, d\phi \, d\rho\right]. \tag{46}$$

The integrals factor because s and (ρ, ϕ) separate:

$$\int_{-L/2}^{L/2} s^2\, ds = \frac{L^3}{12}\,, \quad \int_{-L/2}^{L/2} ds = L\,, \quad \int_0^{\pi}\int_0^{R} \rho\, d\rho\, d\phi = \frac{\pi R^2}{2}\,, \tag{47}$$

$$\int_0^{\pi}\int_0^{R} (\rho\sin\phi - \delta)^2 \rho\, d\rho\, d\phi = \int_0^{\pi}\int_0^{R} (\rho^2\sin^2\phi - 2\delta\rho\sin\phi + \delta^2)\rho\, d\rho\, d\phi \tag{48}$$

$$= \frac{R^4}{4}\cdot\frac{\pi}{2} - 2\delta\cdot\frac{R^3}{3}\cdot 2 + \delta^2\cdot\frac{R^2}{2}\cdot\pi \tag{49}$$

$$= \frac{\pi R^4}{8} - \frac{4\delta R^3}{3} + \frac{\pi\delta^2 R^2}{2}\,. \tag{50}$$

Substituting $\delta = 4R/(3\pi)$:

$$= \frac{\pi R^4}{8} - \frac{4R^3}{3}\cdot\frac{4R}{3\pi} + \frac{\pi R^2}{2}\cdot\frac{16R^2}{9\pi^2} = \frac{\pi R^4}{8} - \frac{16R^4}{9\pi} + \frac{8R^4}{9\pi} = \frac{\pi R^4}{8} - \frac{8R^4}{9\pi}\,. \tag{51}$$

Therefore:

$$I_{ee} = \rho_0\left[\frac{L^3}{12}\cdot\frac{\pi R^2}{2} + L\cdot\left(\frac{\pi R^4}{8} - \frac{8R^4}{9\pi}\right)\right] = \frac{2m}{\pi R^2 L}\left[\frac{\pi R^2 L^3}{24} + L\left(\frac{\pi R^4}{8} - \frac{8R^4}{9\pi}\right)\right] \tag{52}$$

$$= m\left[\frac{L^2}{12} + \frac{R^2}{4} - \frac{16R^2}{9\pi^2}\right] = m\left[\frac{L^2}{12} + \frac{(9\pi^2 - 64)R^2}{36\pi^2}\right]. \checkmark \tag{53}$$

B.2 $I_{bb} = \int (x_b^2 + z_b^2)\, dm$

Since $\int_0^{\pi}\cos^2\phi\, d\phi = \pi/2$:

$$I_{bb} = \rho_0\left[\frac{L^3}{12}\cdot\frac{\pi R^2}{2} + L\cdot\frac{R^4}{4}\cdot\frac{\pi}{2}\right] = m\left[\frac{L^2}{12} + \frac{R^2}{4}\right] = \frac{m(L^2 + 3R^2)}{12}\,. \checkmark \tag{54}$$

B.3 $I_{nn} = \int (x_b^2 + y_b^2)\, dm$

$$I_{nn} = \rho_0\cdot L\cdot\int_0^{\pi}\int_0^{R} [\rho^2\cos^2\phi + (\rho\sin\phi - \delta)^2]\rho\, d\rho\, d\phi = \rho_0 L\left[\frac{\pi R^4}{4} - \frac{8R^4}{9\pi} + \frac{\pi R^4}{8}\right]. \tag{55}$$

Wait—let me redo this carefully:

$$\int_0^{\pi}\int_0^{R} [\rho^2 + \delta^2 - 2\delta\rho\sin\phi]\rho\, d\rho\, d\phi = \frac{R^4}{4}\cdot\pi + \delta^2\cdot\frac{R^2}{2}\cdot\pi - 2\delta\cdot\frac{R^3}{3}\cdot 2 \tag{56}$$

$$= \frac{\pi R^4}{4} + \frac{8R^4}{9\pi} - \frac{16R^4}{9\pi} = \frac{\pi R^4}{4} - \frac{8R^4}{9\pi}\,. \tag{57}$$

So $I_{nn} = \frac{2m}{\pi R^2 L}\cdot L\cdot(\frac{\pi R^4}{4} - \frac{8R^4}{9\pi}) = m(\frac{R^2}{2} - \frac{16R^2}{9\pi^2}) = \frac{(9\pi^2 - 32)mR^2}{18\pi^2}$. ✓

B.4 Off-diagonal elements

$I_{xy} = -\int x_b y_b\, dm$: the integrand $\rho\cos\phi(\rho\sin\phi - \delta) = \rho^2\sin\phi\cos\phi - \delta\rho\cos\phi$. Both $\sin\phi\cos\phi$ and $\cos\phi$ integrate to zero over $[0, \pi]$. So $I_{xy} = 0$. Similarly $I_{xz} = I_{yz} = 0$ (all cross terms involve odd functions).

C Half-Cylinder Angular Momentum Integration

Each half-cylinder extends from $s = 0$ to $s = L$ along its axis from the joint. Mass element position from the system center of mass:

$$\boldsymbol{p}_f = s\,\hat{\boldsymbol{n}}_f + \rho\cos\phi\,\hat{\boldsymbol{e}}_f + \rho\sin\phi\,\hat{\boldsymbol{b}}_f - \boldsymbol{r}_{\text{com}}^{(J)}\,, \tag{58}$$

where $\boldsymbol{r}_{\text{com}}^{(J)} = (0, y_{\text{com}}, 0)$ with $y_{\text{com}} = (L/2)\cos\alpha - \delta\sin\alpha$. The integration is over $\phi \in [0, \pi]$, $\rho \in [0, R]$, $s \in [0, L]$, with density $\rho_0 = 2m/(\pi R^2 L)$.

Since $\hat{\boldsymbol{b}}_f = \hat{\boldsymbol{e}}_{2f}$ and $\hat{\boldsymbol{e}}_f = \hat{\boldsymbol{z}}$, the Cartesian components are identical in form to the full-cylinder case (32)–(34) but with the COM offset changed:

$$p_x = s\sin\alpha + \rho\cos\phi\cos\alpha, \tag{59}$$

$$p_y = s\cos\alpha - \rho\cos\phi\sin\alpha - y_{\text{com}}, \tag{60}$$

$$p_z = \rho\sin\phi. \tag{61}$$

The spin velocity has the same form as (35).

The key difference is that $\int_0^{\pi}\sin\phi\,d\phi = 2 \neq 0$ (unlike $\int_0^{2\pi}\sin\phi\,d\phi = 0$ for the full cylinder). This means terms proportional to $\sin\phi$ in the cross product *survive* the ϕ-integration. However, by the left–right symmetry of the two-cylinder system with the no-wringing constraint, these extra terms cancel between front and rear when computing the total angular momentum, yielding $L_y = L_z = 0$ as in the full-cylinder case.

The calculation was performed symbolically in SymPy, integrating each of the nine components of $\boldsymbol{p} \times \boldsymbol{v}$ (three position components crossed with two velocity contributions, giving $3 \times 3 = 9$ integrands) over all three variables. The result is (19)–(21), verified numerically at $R = 1$, $L = 4$, $\theta = \pi/2$ by checking $L_x = 0$ at the computed gearing ratio to machine precision ($|L_x| < 10^{-16}$).

D Full-Cylinder Torque Derivation

From (41), the front cylinder's y-angular momentum is:

$$(L_f)_y = \frac{m}{24}\big[\Omega_x\sin(2\alpha)(3R^2 - \ell^2) + \Omega_y(\cdots) + 12R^2\omega\cos\alpha\big]. \tag{62}$$

Setting $\Omega_y = 0$ (from $L_y = 0$) and $\Omega_x = G\omega$:

$$(L_f)_y = \frac{m}{24}\big[\sin(2\alpha)(3R^2 - \ell^2)G + 12R^2\cos\alpha\big]\omega. \tag{63}$$

Defining $C^{(\text{full})} = \frac{m}{24}\cdot 12R^2\cos\alpha = \frac{mR^2\cos\alpha}{2}$ and $D^{(\text{full})} = \frac{m}{24}\sin(2\alpha)(3R^2 - \ell^2)$:

$$(L_f)_y = (C^{(\text{full})} + D^{(\text{full})}G)\,\omega. \tag{64}$$

The torque is $\tau_y = d(L_f)_y/dt = (C^{(\text{full})} + D^{(\text{full})}G)\dot{\omega}$.

Physical decomposition:

The spin contribution to $(L_f)_y$ is $(I_{\text{ax}}\omega\hat{\boldsymbol{n}}_f)_y = \frac{mR^2}{2}\omega\cos\alpha = C^{(\text{full})}\omega$.

The whole-body rotation contribution is $(I_{xy}^{(\text{lab})}G\omega)$ where $I_{xy}^{(\text{lab})}$ is the off-diagonal element of the front cylinder's inertia tensor about the system center of mass. From (40), setting $\omega = 0$: $(L_f)_x = \frac{m}{24}[\Omega_x(\cdots) + \Omega_y\sin(2\alpha)(3R^2 - \ell^2)]$, so $I_{xy}^{(\text{lab})} = \frac{m}{24}\sin(2\alpha)(3R^2 - \ell^2) = D^{(\text{full})}$.

Verification that $(L_f)_x = 0$:

From (40): $(L_f)_x = \frac{m}{24}[\Omega_x(2\ell^2\cos^2\alpha - 6R^2\cos^2\alpha + 12R^2) + 12R^2\omega\sin\alpha]$ (with $\Omega_y = 0$).

Substituting $\Omega_x = G\omega$: $(L_f)_x = \frac{m\omega}{24}[G\mathcal{I}_{xx}^{(\text{full})} + 12R^2\sin\alpha]\cdot\frac{1}{2} = \frac{m\omega}{48}\cdot 2[G\mathcal{I}_{xx}^{(\text{full})} + 6R^2\sin\alpha\cdot 2]\ldots$

Actually, note that $(L_f)_x = \frac{1}{2}L_x^{(\text{total})}$ by symmetry (both halves contribute equally to L_x), and $L_x^{(\text{total})} = 0$ by the gearing condition. Therefore $(L_f)_x = 0$. ✓

E Half-Cylinder Torque Derivation

The front half-cylinder's angular momentum has three contributions:

1. Spin angular momentum about its own center of mass: $\boldsymbol{L}_{\text{spin}} = I_{nn}\omega\,\hat{\boldsymbol{n}}_f$. Components:

$$(L_{\text{spin}})_x = I_{nn}\omega\sin\alpha, \qquad (L_{\text{spin}})_y = I_{nn}\omega\cos\alpha. \tag{65}$$

2. Orbital angular momentum from center-of-mass motion: The front half-cylinder's center of mass is at $\boldsymbol{c}_f = (d_x, 0, 0)$ from the system center of mass. Its velocity from the spin is:

$$\boldsymbol{v}_{\text{com}}^{\text{spin}} = \omega\,\hat{\boldsymbol{n}}_f \times (\delta\,\hat{\boldsymbol{b}}_f) = -\omega\delta\,\hat{\boldsymbol{e}}_f = (0, 0, -\omega\delta). \tag{66}$$

(Using $\hat{\boldsymbol{n}}_f \times \hat{\boldsymbol{b}}_f = -\hat{\boldsymbol{e}}_f = -\hat{\boldsymbol{z}}$.)

The orbital angular momentum is:

$$\boldsymbol{L}_{\text{orb}} = m\,\boldsymbol{c}_f \times \boldsymbol{v}_{\text{com}}^{\text{spin}} = m(d_x, 0, 0) \times (0, 0, -\omega\delta) = (0,\ m\,d_x\,\omega\,\delta,\ 0). \tag{67}$$

So $(L_{\text{orb}})_y = m\,d_x\,\delta\,\omega$ with $d_x = (L/2)\sin\alpha + \delta\cos\alpha$.

This term is **zero for full cylinders** ($\delta = 0$).

3. Whole-body rotation: The angular momentum from rotation at Ω_x about $\hat{\boldsymbol{x}}$ is $\boldsymbol{L}_{\text{rot}} = \mathsf{I}_f^{(\text{lab})}\cdot\Omega_x\,\hat{\boldsymbol{x}}$. The relevant components are:

$$(L_{\text{rot}})_x = I_{xx}^{(\text{lab})}\,\Omega_x, \tag{68}$$

$$(L_{\text{rot}})_y = I_{xy}^{(\text{lab})}\,\Omega_x, \tag{69}$$

where $I_{xx}^{(\text{lab})}$ and $I_{xy}^{(\text{lab})}$ are components of the front half-cylinder's inertia tensor about the system center of mass, computed by direct integration (or extracted from the SymPy results as the coefficients of Ω_x in the total angular momentum expressions).

Cancellation of x-components:

$$(L_f)_x = I_{nn}\omega\sin\alpha + I_{xx}^{(\text{lab})}\Omega_x = 0, \tag{70}$$

which gives $I_{xx}^{(\text{lab})}\,G = -I_{nn}\sin\alpha$. This is the gearing condition applied to one half.

Total y-component:

$$(L_f)_y = I_{nn}\omega\cos\alpha + m\,d_x\,\delta\,\omega + I_{xy}^{(\text{lab})}\,G\,\omega = (C^{(\text{half})} + D^{(\text{half})}G)\,\omega, \tag{71}$$

where $C^{(\text{half})} = I_{nn}\cos\alpha + m\,d_x\,\delta$ collects the spin and orbital terms, and $D^{(\text{half})} = I_{xy}^{(\text{lab})}$ is the whole-body rotation contribution.

Explicit evaluation of $C^{(\text{half})}$:

$$C^{(\text{half})} = \frac{(9\pi^2 - 32)mR^2}{18\pi^2}\cos\alpha + m\big(\tfrac{L}{2}\sin\alpha + \tfrac{4R}{3\pi}\cos\alpha\big)\cdot\frac{4R}{3\pi} \tag{72}$$

$$= \frac{(9\pi^2 - 32)mR^2\cos\alpha}{18\pi^2} + \frac{2mLR\sin\alpha}{3\pi} + \frac{16mR^2\cos\alpha}{9\pi^2} \tag{73}$$

$$= mR^2\cos\alpha\left[\frac{9\pi^2 - 32}{18\pi^2} + \frac{16}{9\pi^2}\right] + \frac{2mLR\sin\alpha}{3\pi} \tag{74}$$

$$= mR^2\cos\alpha\cdot\frac{9\pi^2 - 32 + 32}{18\pi^2} + \frac{2mLR\sin\alpha}{3\pi} \tag{75}$$

$$= \frac{mR^2\cos\alpha}{2} + \frac{2mLR\sin\alpha}{3\pi} \tag{76}$$

$$= \frac{mR(3\pi R\cos\alpha + 4L\sin\alpha)}{6\pi}\,. \checkmark \tag{77}$$

Note the remarkable simplification: $I_{nn}\cos\alpha + m\delta^2\cos\alpha = (mR^2/2)\cos\alpha$, recovering the *full-cylinder* axial moment times $\cos\alpha$. The additional $2mLR\sin\alpha/(3\pi)$ term is purely from the orbital contribution's L-dependence.

Numerical verification at $R = 1$, $L = 4$, $\theta = 90°$:

$$I_{nn}\cos\alpha = 0.3199 \times 0.7071 = 0.2262, \tag{78}$$

$$m\,d_x\,\delta = 1 \times (2 \times 0.7071 + 0.4244 \times 0.7071) \times 0.4244 = 0.7276, \tag{79}$$

$$C^{(\text{half})} = 0.2262 + 0.7276 = 0.9538. \tag{80}$$

Cross-check: $mR(3\pi R\cos\alpha + 4L\sin\alpha)/(6\pi) = 1 \times 1 \times (3\pi \times 0.7071 + 4 \times 4 \times 0.7071)/(6\pi) = (6.664 + 11.314)/18.850 = 0.9538$. ✓

The torque on the front half is therefore:

$$\boldsymbol{\tau}_f^{(\text{half})} = (C^{(\text{half})} + D^{(\text{half})}G^{(\text{half})})\,\dot{\omega}\,\hat{\boldsymbol{y}}\,. \tag{81}$$

At $R = 1$, $L = 4$, $\theta = 90°$: $C^{(\text{half})} + D^{(\text{half})}G^{(\text{half})} = 0.954 + (-0.632)(-0.238) = 0.954 + 0.150 = 1.104$. The torque on the rear half is the negative of this, along $-\hat{\boldsymbol{y}}$.